Image Processing and Medical Imaging

Feature Extraction

This work was supported by the construction EFOP-3.4.3-16-2016-00021. The project was supported by the European Union, co-financed by the European Social Fund.

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Shape representation and description

- The last talk was devoted to image segmentation methods and showed how to construct homogeneous regions.
- Important step for exact **region description**.
- This description should generate a numeric feature vector.
- Defining the shape of an object can prove to be very difficult.
- Verbally we can use terms such as **elongated**, **rounded**, whit **sharp edges**, etc.

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Region identification

- Region identification is necessary for region description.
- One of the many methods for region identification is to label each region with a unique (integer) number.
- This is so-called labeling or **connected component labeling**.

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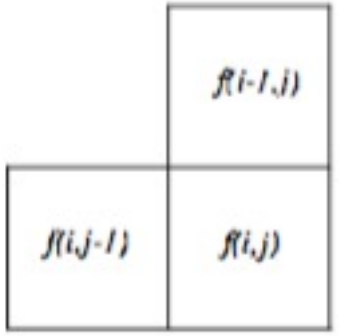
Algorithm for region identification (1st pass)

- Search the image *R* row by row and assign a non-zero value *v* to each non-zero pixel *R(i,j)*.
- The value *v* is chosen according to the labels of pixel's neighbors, where the neighborhood is defined by



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Algorithm for region identification (1st pass)

- If all the neighbors are background pixels, *R(i,j)* is assigned a new unused label.
- If there is just one neighboring pixel with a non-zero label, assign this label to the pixel *R(i,j)*.
- If there is more than one non-zero pixel among the neighbors, assign the label of any one to the labeled pixel. If the labels of any of the neighbors differ (label collision), store the label pair as being equivalent.

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Algorithm for region identification (2nd pass)

- Because some regions have pixels with different labels. The whole image is scanned again, and pixels are re-labeled using the stored information.

1			2
1			2
1			2
1	1	1	?

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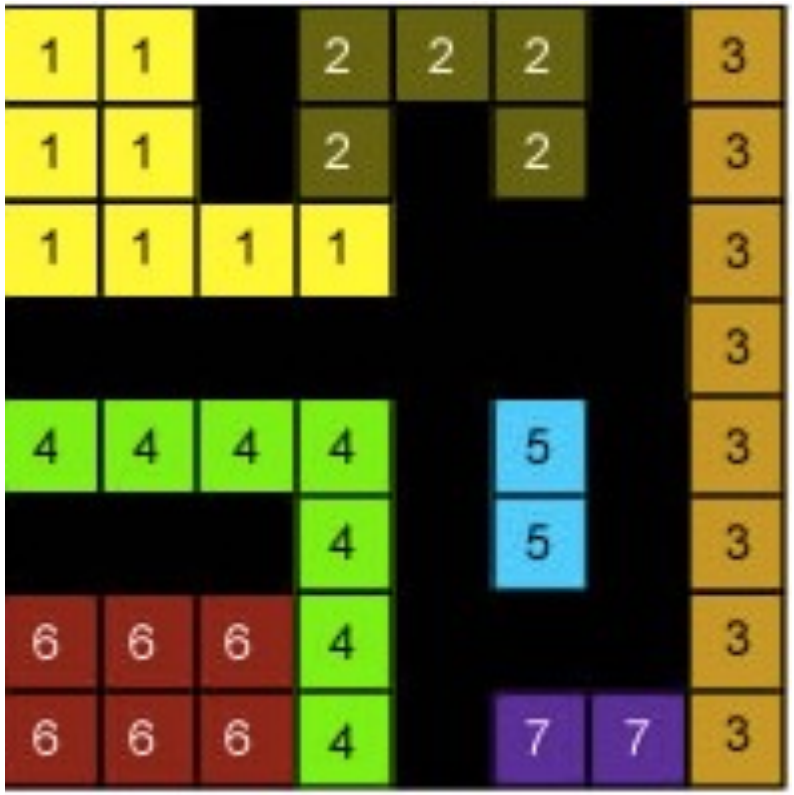
Algorithm for region identification (Example)



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Algorithm for region identification (Example)

1	1		2	2	2		3
1	1		2		2		3
1	1	1	1		2		3
							3
4	4	4	4		5		3
			4		5		3
6	6		4				3
6	6		4		7	7	3

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Algorithm for region identification (Example)

1	1		1	1	1		3
1	1		1		1		3
1	1	1	1		1		3
							3
4	4	4	4		5		3
			4		5		3
6	6		4				3
6	6		4		3	3	3

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Simple geometric border representation

- **Boandry length.** Vertical and horizontal steps have unit length, and the length of diagonal steps is √2.
- **Curvature.** Ratio between the total number of boundary pixels and the number of boundary pixels where the boundary direction changes significantly.
- **Bending energy.** It can be computed as a sum of squares of the border curvate *c(k)* over the border length *L.*

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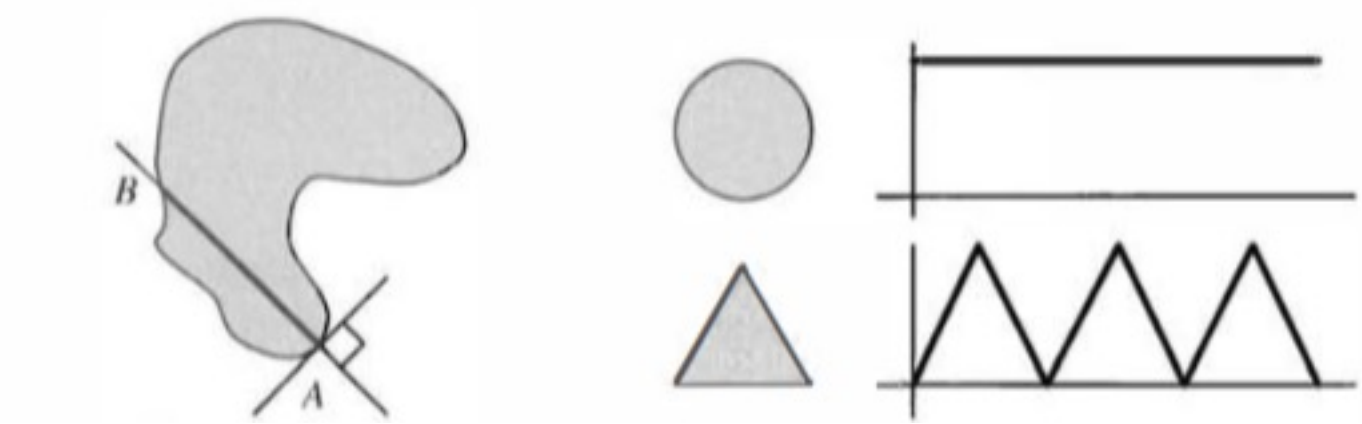
Simple geometric border representation (cont.)

- **Signature.**



Source: http://user.engineering.uiowa.edu/~dip/Lecture/Shape2.html

Simple geometric border representation (cont.)

- Fourier transform of boundaries
- Polygonal representation of boundaries
- B-spline representation of boundaries
- Hough transform

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Simple scalar region descriptors

- Area
- Euler's number
- Projections
- Eccentricity
- Elongatedness (*area*/2x(*erosion-steps*)^2)
- Direction
- Compactness (perimeter^2/area)
- Convex hull

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Graph representation based on region skeleton

Source: https://en.wikipedia.org/wiki/Topological_skeleton

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Hough transform

- The simplest case of Hough transform is detecting straight lines.
- In general, the straight line $y=mx+b$ can be represented as a point (b,m) in the parameter space.
- However, vertical line pose a problem.
- Given a single point in the plane, then the set of all straight lines going through that point corresponds to straight line in the parameter plane.
- Thus, the problem of detecting collinear points can be converted to the problem of finding concurrent lines.

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Hough transform (cont)

https://www.youtube.com/watch?v=4zHbI-fFIlI

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Walsh transform in 1D

- The Walsh transform and its inverse transform in one-dimensional space can be defined in the following form:

$$W(\mathrm{u})=\frac{1}{N}\sum_{x=0}^{N-1} f(x)\prod_{i=0}^{n-1}(-1)^{b_i(x)b_{n-1-i}(u)}$$

$$f(\mathrm{x})=\sum_{u=0}^{N-1} W(u)\prod_{i=0}^{n-1}(-1)^{b_i(x)b_{n-1-i}(u)}$$

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Walsh transform in 2D

$$W(\mathrm{u,v})=\frac{1}{N}\sum_{x=0}^{N-1}\sum_{y=0}^{N-1} f(x,y)\prod_{i=0}^{n-1}(-1)^{(b_i(x)b_{n-1-i}(u)+(b_i(y)b_{n-1-i}(v))}$$

$$f(\mathrm{x,y})=\frac{1}{N}\sum_{u=0}^{N-1}\sum_{v=0}^{N-1} W(u,v)\prod_{i=0}^{n-1}(-1)^{(b_i(x)b_{n-1-i}(u)+(b_i(y)b_{n-1-i}(v))}$$

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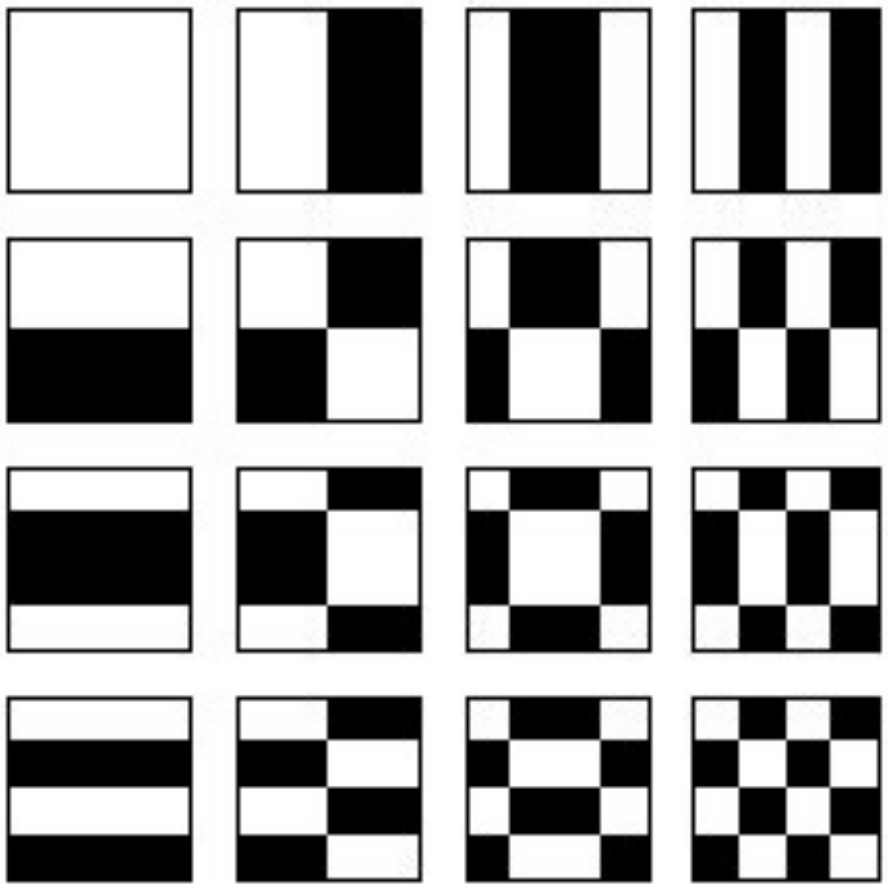
Basis images



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Thank you for your attention